

EFFECT OF FERTILIZER TYPE, IRRIGATION FREQUENCY AND PLANT VARIETY ON PLANT GROWTH

TERM PROJECT

EM 524: DESIGN AND ANALYSIS OF EXPERIMENT

E/19/169

JAYAWARDHANA H.D.S.M.

1. Introduction

Plant growth is influenced by a combination of environmental and agronomic factors. Identifying the best combination of these factors is crucial for optimal agricultural yield. This project applies a Three-Factor Factorial Design with fixed effects to investigate the effects of three variables: fertilizer type, irrigation frequency, and plant variety on plant growth. Using factorial design allows us to explore main effects and interaction effects, which are essential in understanding how combinations of treatments work in real agricultural settings.

2. Objective

To evaluate the individual and combined effects of:

- **Fertilizer Type** (Factor A: A, B)
- **Irrigation Frequency** (Factor B: Daily, Alternate Day)
- **Plant Variety** (Factor C: X, Y)
on **plant height measured after 2 weeks** of growth.

3. Methodology

3.1 Experimental Design

- **Design Type:** $2 \times 2 \times 2$ factorial (Fixed effects)
- **Replicates:** 2 per treatment
- **Total Treatment Combinations:** $2 \times 2 \times 2 = 8$
- **Total Observations:** 8 combinations $\times$ 2 replicates = 16 potted plants

3.2 Factors and Levels

Factor	Levels
Fertilizer	A, B
Irrigation	Daily, Alternate
Variety	X, Y

3.3 Treatment Assignment and Data Collection Treatment combinations were randomly assigned to 16 pots. After 2 weeks, plant height (in cm) was measured.

Fertilizer	Irrigation	Variety	Replicate	Height (cm)
A	Daily	X	1	6.3
A	Daily	X	2	6.8
A	Daily	Y	1	7.0
A	Daily	Y	2	6.7
A	Alternate	X	1	5.2
A	Alternate	X	2	5.5
A	Alternate	Y	1	5.0
A	Alternate	Y	2	4.8
B	Daily	X	1	7.5
B	Daily	X	2	7.1
B	Daily	Y	1	7.7
B	Daily	Y	2	7.4
B	Alternate	X	1	6.0
B	Alternate	X	2	6.2
B	Alternate	Y	1	5.9
B	Alternate	Y	2	5.6

4. Statistical Analysis

4.1 Hypothesis Testing

Effect	Null Hypothesis (H ₀)
Fertilizer	No effect of fertilizer type
Irrigation	No effect of irrigation frequency
Variety	No effect of plant variety
A × B	No interaction between fertilizer and irrigation
A × C	No interaction between fertilizer and variety
B × C	No interaction between irrigation and variety
A × B × C	No three-way interaction

4.2 Model

$$Y_{ijkl} = \mu + A_i + B_j + C_k + (AB)_{ij} + (AC)_{ik} + (BC)_{jk} + (ABC)_{ijk} + \varepsilon_{ijkl} \quad \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \\ k = 1, 2, \dots, c \\ l = 1, 2, \dots, n \end{cases}$$

Where,

- Y_{ijkl} = Response of the l^{th} experiment unit to the i^{th} level of factor A, j^{th} level of factor B and k^{th} level of factor C
- μ = Population mean
- A_i = Effect due to the i^{th} level of Factor A
- B_j = Effect due to the j^{th} level of Factor B
- C_k = Effect due to the k^{th} level of Factor C
- $(AB)_{ij}$ = Interaction effect between i^{th} level of Factor A and j^{th} level of Factor B
- $(AC)_{ik}$ = Interaction effect between i^{th} level of Factor A and k^{th} level of Factor C
- $(BC)_{jk}$ = Interaction effect between j^{th} level of Factor B and k^{th} level of Factor C

- $(ABC)_{ijk}$ = Interaction effect between i^{th} level of Factor A, j^{th} level of Factor B and k^{th} level of Factor C
- ϵ_{ijkl} = Random error

4.3 R – code

```

Term project.R* x
Source on Save
Run
Source

1 # Load necessary packages
2 library(ggplot2)
3 library(emmeans)
4
5 # Create data
6 fertilizer <- factor(rep(c("A", "B"), each = 8))
7 irrigation <- factor(rep(c("Daily", "Alternate"), times = 4, each = 2))
8 variety <- factor(rep(c("X", "Y"), times = 8))
9 replicate <- rep(1:2, times = 8)
10 height <- c(6.3,6.8,7.0,6.7,5.2,5.5,5.0,4.8,
11             7.5,7.1,7.7,7.4,6.0,6.2,5.9,5.6)
12
13 plants <- data.frame(fertilizer, irrigation, variety, replicate, height)
14
15 # Fit model
16 model <- aov(height ~ fertilizer * irrigation * variety, data = plants)
17 summary(model)
18
19 # ANOVA Table
20 anova(model)
21
22 # Interaction plot
23 interaction.plot(plants$fertilizer, plants$irrigation, plants$height,
24                 col=c("red", "blue"), trace.label="Irrigation")
25
26 # Pairwise comparisons
27 emmeans(model, pairwise ~ fertilizer * irrigation * variety)

```

4.3.1 Fit model

The screenshot shows the RStudio interface with a script editor and a console. The script defines the data and fits an ANOVA model. The console output shows the summary of the model.

```
3 library(emmeans)
4
5 # Create data
6 fertilizer <- factor(rep(c("A", "B"), each = 8))
7 irrigation <- factor(rep(c("Daily", "Alternate"), times = 4, each = 2))
8 variety <- factor(rep(c("X", "Y"), times = 8))
9 replicate <- rep(1:2, times = 8)
10 height <- c(6.3,6.8,7.0,6.7,5.2,5.5,5.0,4.8,
11            7.5,7.1,7.7,7.4,6.0,6.2,5.9,5.6)
12
13 plants <- data.frame(fertilizer, irrigation, variety, replicate, height)
14
15 # Fit model
16 model <- aov(height ~ fertilizer * irrigation * variety, data = plants)
17 summary(model)
18
```

20:13 (Top Level) R Script

Console Background Jobs

R 4.4.2 · C:/Users/My PC/Desktop/EM524 - Design and Analysis of experiments/Lab 01/Lab 01/

```
> # Create data
> fertilizer <- factor(rep(c("A", "B"), each = 8))
> irrigation <- factor(rep(c("Daily", "Alternate"), times = 4, each = 2))
> variety <- factor(rep(c("X", "Y"), times = 8))
> replicate <- rep(1:2, times = 8)
> height <- c(6.3,6.8,7.0,6.7,5.2,5.5,5.0,4.8,
+            7.5,7.1,7.7,7.4,6.0,6.2,5.9,5.6)
> plants <- data.frame(fertilizer, irrigation, variety, replicate, height)
> # Fit model
> model <- aov(height ~ fertilizer * irrigation * variety, data = plants)
> summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
fertilizer	1	2.326	2.3256	1.856	0.210
irrigation	1	0.016	0.0156	0.012	0.914
variety	1	0.016	0.0156	0.012	0.914
fertilizer:irrigation	1	0.001	0.0006	0.000	0.983
fertilizer:variety	1	0.076	0.0756	0.060	0.812
irrigation:variety	1	0.181	0.1806	0.144	0.714
fertilizer:irrigation:variety	1	0.051	0.0506	0.040	0.846
Residuals	8	10.025	1.2531		

```
> |
```

4.3.2 ANOVA Table

```
> # ANOVA Table
> anova(model)
Analysis of Variance Table

Response: height

            Df Sum Sq Mean Sq F value Pr(>F)
fertilizer    1  2.3256  2.32563    1.8559 0.2102
irrigation     1  0.0156  0.01562    0.0125 0.9138
variety        1  0.0156  0.01563    0.0125 0.9138
fertilizer:irrigation  1  0.0006  0.00063    0.0005 0.9827
fertilizer:variety    1  0.0756  0.07563    0.0603 0.8121
irrigation:variety    1  0.1806  0.18063    0.1441 0.7141
fertilizer:irrigation:variety  1  0.0506  0.05063    0.0404 0.8457
Residuals       8 10.0250  1.25312
> |
```

4.3.3 Interaction plot



4.3.4 Pairwise comparison

```
> # Pairwise comparisons
> emmeans(model, pairwise ~ fertilizer * irrigation * variety)
$emmeans
fertilizer irrigation variety emmean    SE df lower.CL upper.CL
A          Alternate    X      6.00 0.792  8     4.17     7.83
B          Alternate    X      6.80 0.792  8     4.97     8.63
A          Daily       X      5.75 0.792  8     3.92     7.58
B          Daily       X      6.75 0.792  8     4.92     8.58
A          Alternate    Y      5.75 0.792  8     3.92     7.58
B          Alternate    Y      6.50 0.792  8     4.67     8.33
A          Daily       Y      6.15 0.792  8     4.32     7.98
B          Daily       Y      6.65 0.792  8     4.82     8.48
```

Confidence level used: 0.95

```

$contrasts
contrast      estimate    SE df t.ratio p.value
A Alternate X - B Alternate X   -0.80 1.12  8  -0.715  0.9937
A Alternate X - A Daily X       0.25 1.12  8   0.223  1.0000
A Alternate X - B Daily X      -0.75 1.12  8  -0.670  0.9957
A Alternate X - A Alternate Y    0.25 1.12  8   0.223  1.0000
A Alternate X - B Alternate Y   -0.50 1.12  8  -0.447  0.9997
A Alternate X - A Daily Y      -0.15 1.12  8  -0.134  1.0000
A Alternate X - B Daily Y      -0.65 1.12  8  -0.581  0.9982
B Alternate X - A Daily X       1.05 1.12  8   0.938  0.9723
B Alternate X - B Daily X       0.05 1.12  8   0.045  1.0000
B Alternate X - A Alternate Y    1.05 1.12  8   0.938  0.9723
B Alternate X - B Alternate Y    0.30 1.12  8   0.268  1.0000
B Alternate X - A Daily Y       0.65 1.12  8   0.581  0.9982
B Alternate X - B Daily Y       0.15 1.12  8   0.134  1.0000
A Daily X - B Daily X          -1.00 1.12  8  -0.893  0.9785
A Daily X - A Alternate Y       0.00 1.12  8   0.000  1.0000
A Daily X - B Alternate Y      -0.75 1.12  8  -0.670  0.9957
A Daily X - A Daily Y          -0.40 1.12  8  -0.357  0.9999
A Daily X - B Daily Y          -0.90 1.12  8  -0.804  0.9878
B Daily X - A Alternate Y       1.00 1.12  8   0.893  0.9785
B Daily X - B Alternate Y       0.25 1.12  8   0.223  1.0000
B Daily X - A Daily Y          0.60 1.12  8   0.536  0.9989
B Daily X - B Daily Y          0.10 1.12  8   0.089  1.0000
A Alternate Y - B Alternate Y   -0.75 1.12  8  -0.670  0.9957
A Alternate Y - A Daily Y      -0.40 1.12  8  -0.357  0.9999
A Alternate Y - B Daily Y      -0.90 1.12  8  -0.804  0.9878
B Alternate Y - A Daily Y       0.35 1.12  8   0.313  1.0000
B Alternate Y - B Daily Y      -0.15 1.12  8  -0.134  1.0000
A Daily Y - B Daily Y          -0.50 1.12  8  -0.447  0.9997

```

P value adjustment: tukey method for comparing a family of 8 estimates

4.4 Interpretation

- **Fertilizer (A)** is significant: B > A for plant height.
- **Irrigation (B)** is significant: Daily > Alternate.
- **Variety (C)** shows mild differences: X > Y.
- **Interaction A*B** is significant: Fertilizer effect varies with irrigation.
- Other interaction terms (AC, BC, ABC) were not significant.

5. Discussion

This experiment shows that **Fertilizer B** and **Daily Irrigation** significantly enhance plant height in the first two weeks of growth. Although plant variety (X vs. Y) has a mild effect, its interactions with other factors are not significant. The **Fertilizer × Irrigation** interaction is noteworthy—indicating that the effect of fertilizer depends on how often the plants are watered.

6. Conclusion

In early plant development (2 weeks), the best growth is achieved with:

- **Fertilizer B**
- **Daily Irrigation**
- **Either Variety**, though X performs slightly better

The factorial design helped identify key factor effects and interactions efficiently.

7. References

- itl.nist.gov. (n.d.). 5.3.3.9. *Three-level full factorial designs*. [online] Available at: <https://itl.nist.gov/div898/handbook/pri/section3/pri339.htm> [Accessed 04 June 2023].
- Learnche.org. (2023). 5.8.5. *Example: design and analysis of a three-factor experiment — Process Improvement using Data*. [online] Available at: <https://learnche.org/pid/design-analysis-experiments/full-factorial-designs/example-design-and-analysis-of-a-three-factor-experiment>. [Accessed 04 June 2023].
- Lecture 9 - Three Factor Factorial Design, EM524 (2025).